PANCHIP

PAN107 系列

产品说明书

V1.9 Mar. 2025

上海磐启微电子有限公司

BLE SoC 收发器

Overview

The PAN107 series integrates BLE5.3 and 2.4GHz dual-mode wireless transceiver circuit SOC chips. The wireless transceiver circuit operates in the 2.400 ~ 2.483GHz universal ISM frequency band. The PAN107 series has built-in 512KB Flash program memory and 48KB SRAM memory. In addition, the PAN107 series is equipped with a wealth of peripherals, including up to 26 GPIOs, 8 PWMs, 1 32-bit timer, 2 24-bit timers, 1 I2C, 2 UARTs, 2 SPIs, 7 external channels of ADC, WDT, WWDT, USB2.0 (Full speed), 32K RC automatic calibration, etc.
The PAN107 series is suitable for applications such as high-precision indoor positioning (AoA), wireless mouse and keyboard, smart home, and electronic shelf labels.

Key Features

ÿ **MCU** 32-
bit MCU, maximum frequency 48MHz ÿ
Memory
Flash: built-in 512KB, support deep sleep mode SRAM:
48KB
eFuse: 128B
Cache: 4KB ÿLow
power
reception mode: 2.5mA@1Mbps DCDC
transmission mode: 5.06mA@0dBm (DCDC)
standby mode: 0.28uA
standby mode (SRAM retention):
1.88uA (support GPIO / XTL / RCL wake-up) Deep
sleep mode: 3.37uA (All Logic Retention,
GPIO, XTL, RCL can be woken
up) ÿ Clock
32MHz RC 32MHz
XTAL 32kHz RC
32.768kHz XTAL
DPLL(48MHz) ÿ **RF** Support
Modesÿ Various modes
of
BLE5.3: 1Mbps,
2Mbps, 500kbps, 125kbps ÿ
2.4G private protocol: 1Mbps, 2Mbps,
500kbps, 250kbps,
125kbps, support hardware ACK Transmit Power: Up to 9dBm
Receive
Sensitivityÿ -99dBm@125kbps ÿ
-99dBm@500kbps
ÿ -96dBm@1Mbps ÿ
-93dBm@2Mbps RSSI ÿ
Resolution: 0.25dB ÿ
Accuracy: ±2dB ÿ Range:
-90dbm ~
-15dBm Support single
antenna Safety
regulations: BQB / ETSI / FCC

Broadcast and scan once per second, traversing three channels
37, 38, and 39 each time, with an
average
power consumption of
about 13uA ÿ
Peripherals ÿ Up to 26 GPIOs 8 PWMs 1 32-bit
timer, 2 24-bit
timers 1 I2C 2
UARTs 2 SPIs
DMA
12 ADCs (7 ext, VBG_1P2, VBG_VT,
1/4VDD, VBG_0P6, Temp2)
Support WDT / WWDT
Support IO / BOD / POR / LVR / System reset FMC
(support IAP, support bootloader at address 0x0)
sequence) ÿ Clock
measurement, clock calibration
USB2.0 (Full_speed)
Flash data
encryption ÿ Temperature
sensor support Temperature sensor
detection range:
-40ÿ ~ 85ÿ ÿ Power
management integrated voltage regulator Operating
voltage:
1.8V ~3.6V (support DCDC) ÿ Package
QFN32 (4*4) / QFN20 (3*3)
QFN40 (5*5) /
LGA20 ÿ Working conditions Operating temperature:
-40ÿ ~ 85ÿ / -40ÿ ~ 105ÿ Storage temperature: -60ÿ~150ÿ

Typical Applications

ÿ Electronic shelf labels ÿ
Wireless mouse and keyboard
ÿ LED light control

Bluetooth Features

Bluetooth Low Energy ControllersThe

PAN107 series of Bluetooth Low Energy controllers support all low energy features required by Bluetooth specification version 5.3. The controllers support the following features:

ÿSupport **1M PHY, 2M PHY** and Coded **PHY (s2** and **s8)**

Supports Advertising, Scanning, Initiating **and Connection (Central** and **Periph-eral)**

ÿThere are up to **6** link layer state machines at the same time:

Passive Scan × 1 Non-

connectable Broadcast × 1 Any

other combination (traditional/extended/periodic broadcast, scan and connect)

ÿSupports

LE features: LL Encryption

LE Data Packet

Length Extension LL Privacy

Extension

Scanner Filter Policy LE Extension

and Periodic Broadcast Frequency

Hopping Algorithm

#2 Constant Frequency Extension Signal

(CTE) ÿSupports updating channel statistics

Bluetooth Host

ÿ Generic access profile with all possible **LE** roles **(GAP)**

Peripheral & Central

Observer & Broadcaster

ÿ GATT (Generic Attribute Profile) Server

(acting as a sensor) Client

(connecting to a sensor)

ÿ Pairing support, including Bluetooth **4.2** 's Secure Connections feature ÿ

Non-volatile storage supports permanent storage of Bluetooth-specific settings and data

HCI Driver Abstraction

3-Wire (H5) & 5-Wire (H4) UART SPI

Local controller support as virtual HCI driver

Bluetooth **Mesh**

ÿCompatible with Bluetooth **SIG Mesh Profile 1.0.1**

ÿSupport **Mesh** Provisioning

ÿ **Provisioner: PB-ADV**

ÿ **Provisionee: PB-ADV, PB-GATT ,** and **PB-Remote**

ÿSupport **mesh node functions: Relay, Proxy,** Friend, Low Power Node (LPN) ÿSupport mesh model

SIG models: Configuration model, health model, and common model

(switch and light control

model) SIG development models: PB-Remote model and SIG OTA model

type

ÿSupport Baidu Xiaodu, Alibaba **Aligenie, Amazon Echo** and multiple smart speakers to control at the

same time ÿSupport **network control: HeartBeat, Subnet, Secure Beacon** and **Group Control**

ÿSupports non-delayed switching control for more than **256**

nodes ÿMesh

security Provisioning: FIPS P-256 Elliptic Curve Encryption

Message: AES-CCM Encryption Network:

SEQ Control, IV Index and Key Fresh

Private **2.4G** function

ÿ Support 250K, 1M and 2M PHY ÿ

XN297L, PAN1026 transceiver protocol compatible ÿ

Support No Acknowledge, Acknowledge and Acknowledge with Payload ÿ Support

CRC8, CRC16 and CRC24 ÿ Support Whitening

ÿ Compatible with

Bluetooth framework structure, support Bluetooth broadcast and

scan ÿ Compatible with Bluetooth CODED PHY

S2/S8 ÿ Support 2-byte

address ÿ Support the same spread spectrum function as BLE protocol

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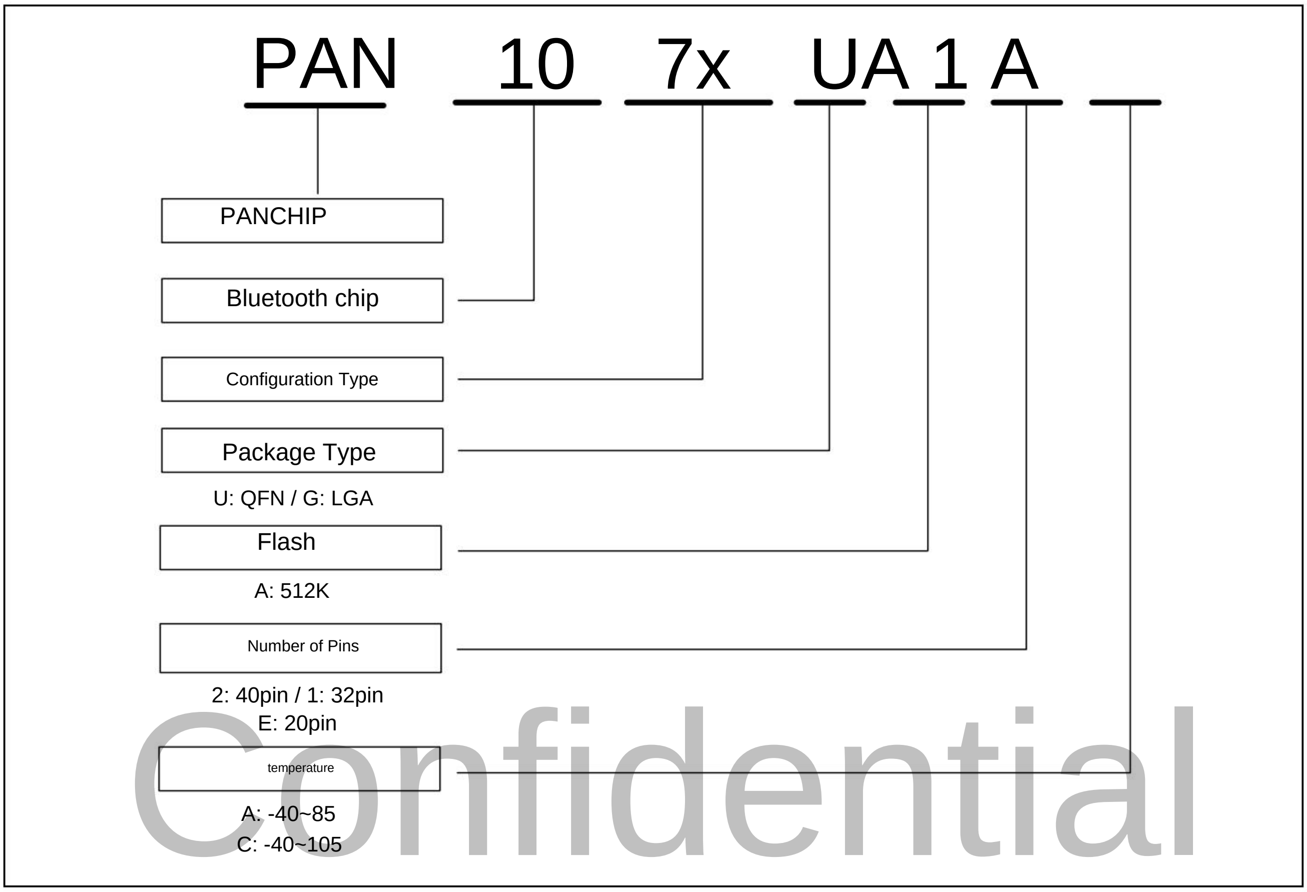
1 Naming Rules



2Ordering Information

Product Model	Chip Type	Package	Pin Number	IO	FLASH	RAM	Temperature	Package
PAN1070UA2A	BLE5.3	QFN	40	26	512K	48K	-40~85ÿ	Tape & Reel
PAN1070UA1A	BLE5.3	QFN	32	twenty one	512K	48K	-40~85ÿ	Tape & Reel
PAN1070UA1C	BLE5.3	QFN	32	twenty one	512K	48K	-40~105ÿ	Tape & Reel
PAN1070UAEC	BLE5.3	QFN	20	8	512K	48K	-40~105ÿ	Tape & Reel
PAN1070GAEC	BLE5.3	LGA	20	10	512K	48K	-40~105ÿ	Tape & Reel

Before ordering, please consult sales for the latest mass production information.

3 System structure block diagram

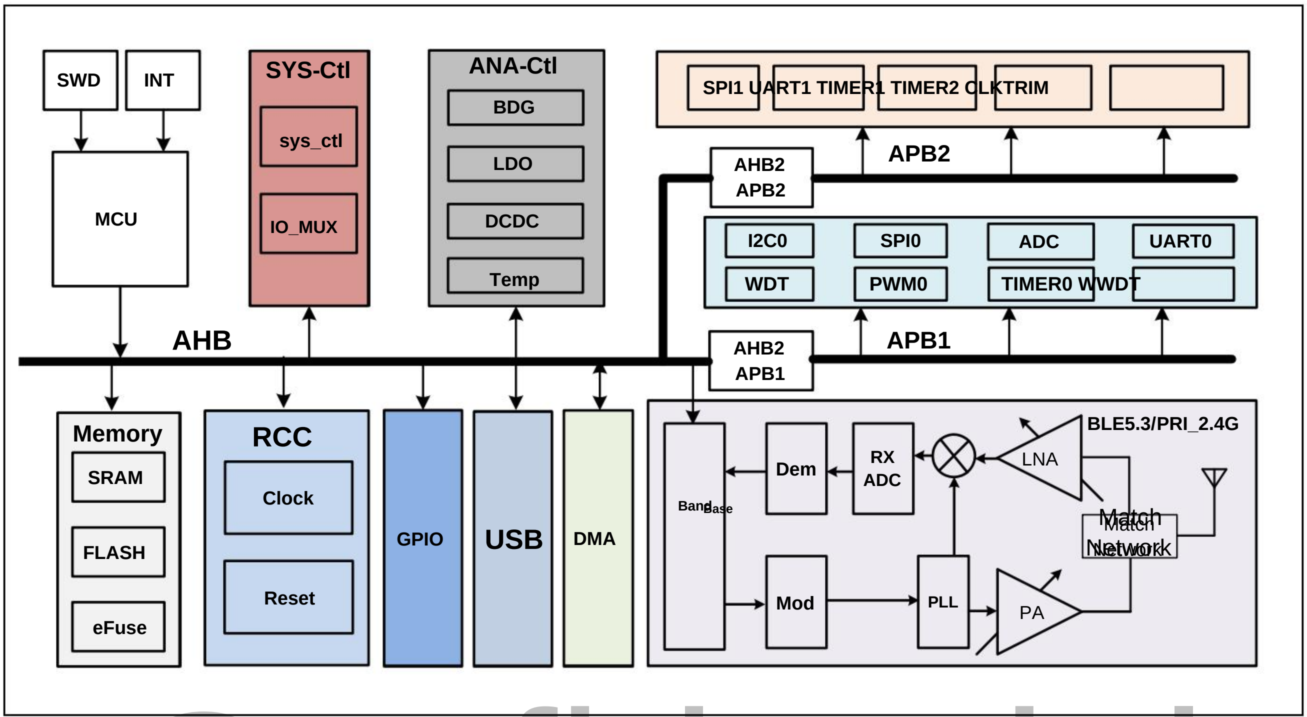


Figure 3-1 System structure block diagram

4 Pin Information

4.1 Pin Diagram

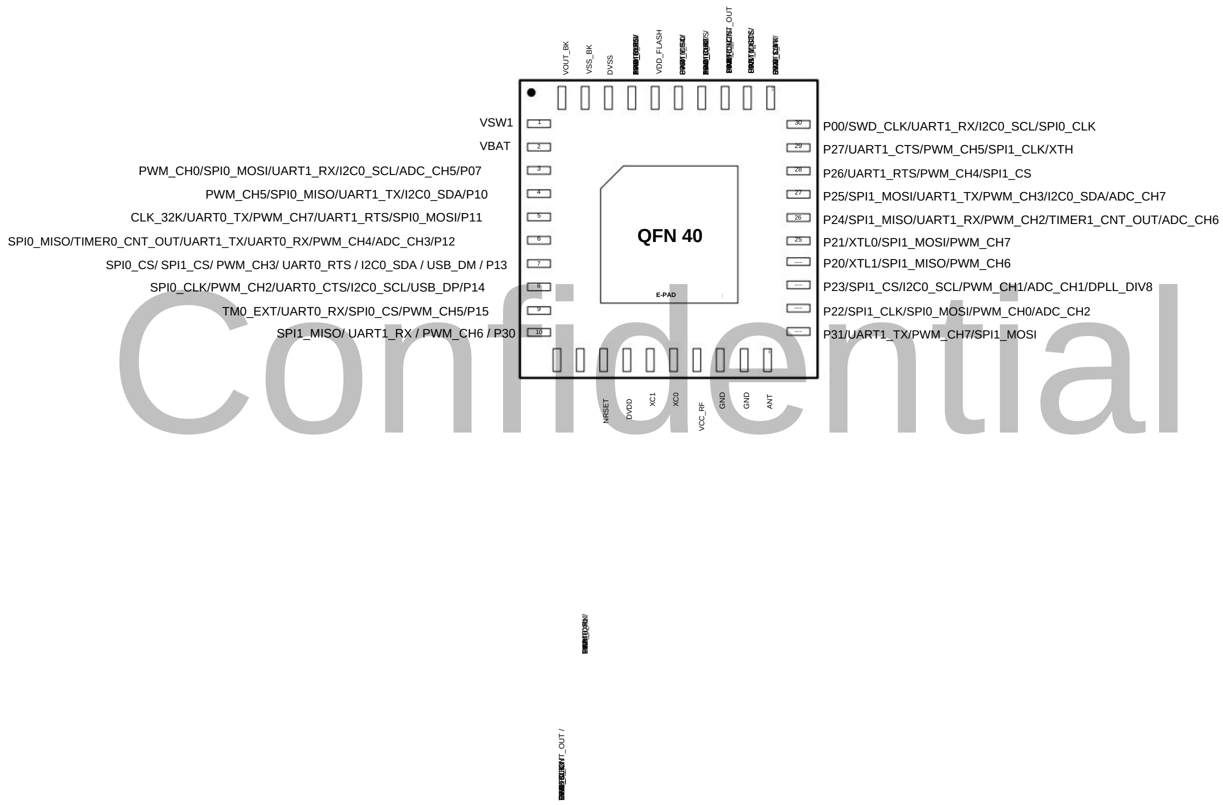


Figure 4-1 QFN 40 pin diagram

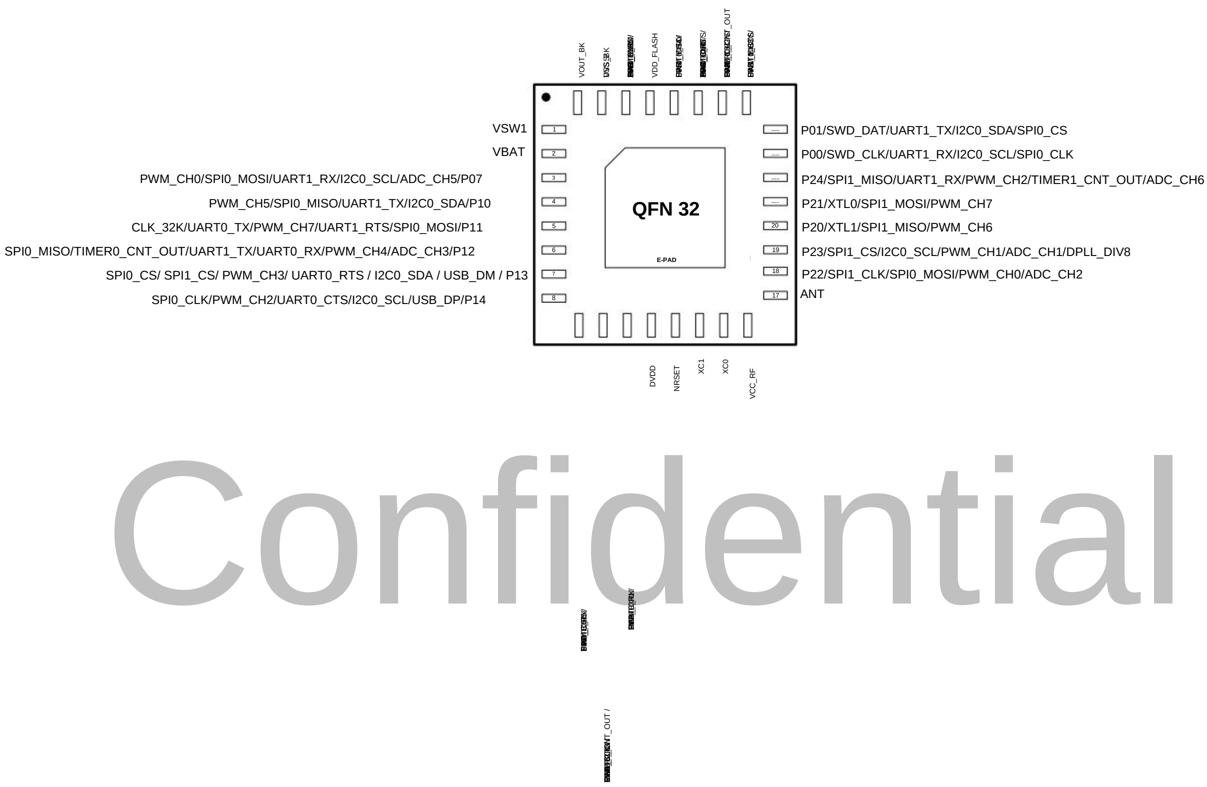


Figure 4-2 QFN 32 pin diagram

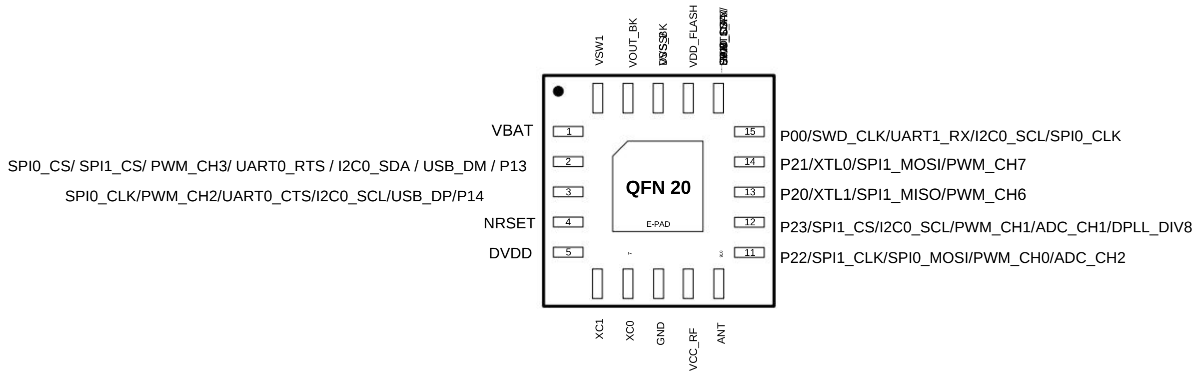


Figure 4-3 QFN 20 pin diagram

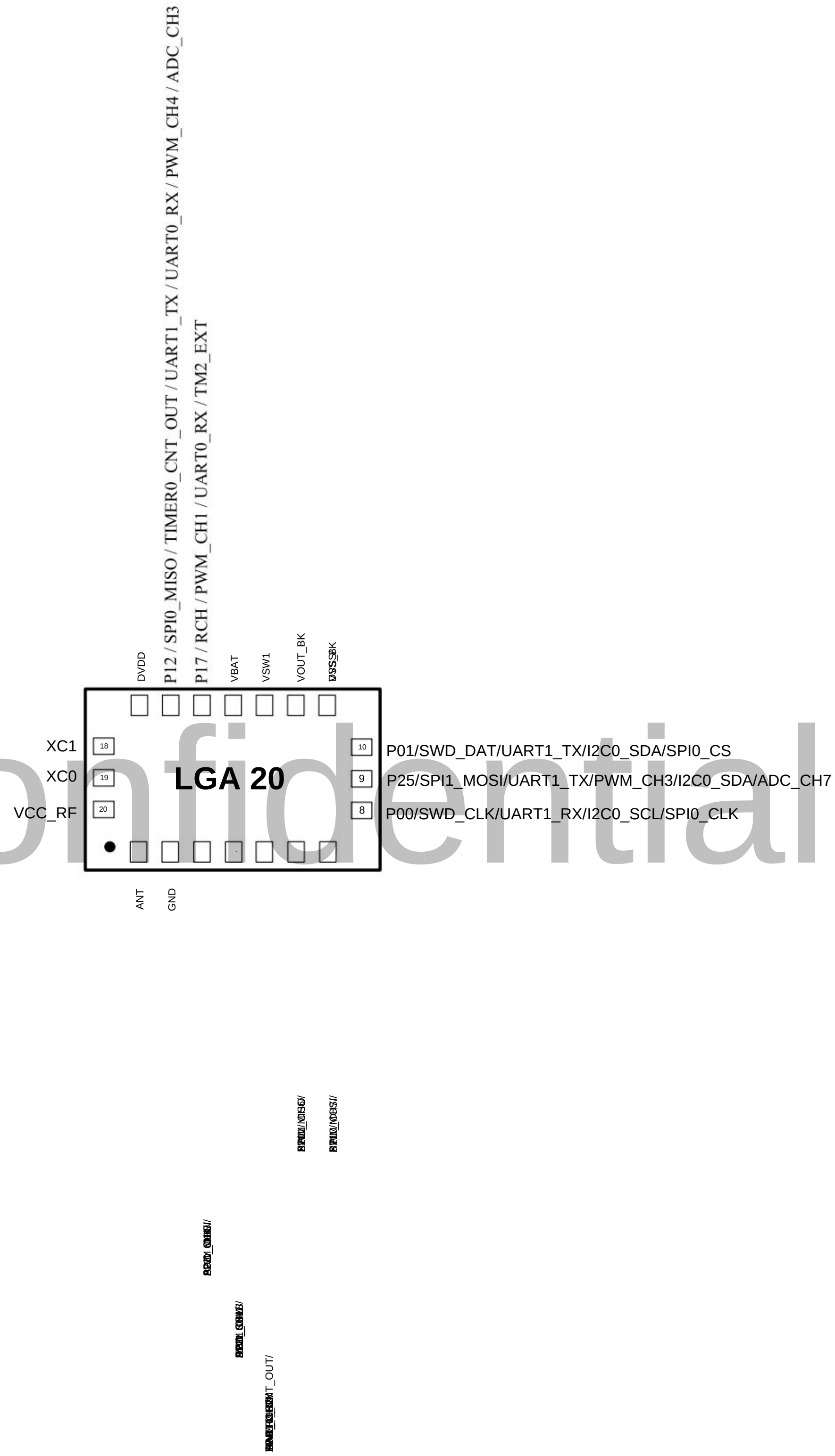


Figure 4-4 LGA 20 pin diagram

4.2 Pin Description

Table 4-1 Pin Description

Encapsulation				Pin Name	Pinout type	describe
QFN40	QFN32	QFN20	LGA20			
1	1	20	13	VSW1	P	DCDC internal power switch (switching frequency about 650kHz), using External inductor is required
2	2	1	14	VBAT	P	chip power input pin (VDD)
3	3	-	-	P07	I/O	General purpose digital input and output pins
				UART1_RX	I	UART1 RX pin
				I2C0_SCL	I/O	I2C0 SCL pin
				SPI0_MOSI	I/O	SPI0 MOSI pin
				PWM_CH0	O	PWM channel 0 output pin
				ADC_CH5	AI	ADC Channel 5 Pin
4	4	-	-	P10	I/O	General purpose digital input and output pins
				UART1_TX	O	UART1 TX pin
				I2C0_SDA	I/O	I2C0 SDA pin
				SPI0_MISO	I/O	SPI0 MISO pin
				PWM_CH5	O	PWM channel 5 output pin
5	5	-	-	P11	I/O	General purpose digital input and output pins
				UART1_RTS	O	UART1 RTS pin
				SPI0_MOSI	I/O	SPI0 MOSI pin
				PWM_CH7	O	PWM channel 7 output pin
				CLK_32K	O	CLK_32K output pin
				UART0_TX	O	UART0 TX pin
6	6	-	16	P12	I/O	General purpose digital input and output pins
				UART0_RX	I	UART0 RX pin
				TIMER0_CNT_OUT	O	TIMER0 output pin
				UART1_TX	O	UART1 TX pin
				SPI0_MISO	I/O	SPI0 MISO pin
				PWM_CH4	O	PWM channel 4 output pin
				ADC_CH3	AI	ADC Channel 3 Pin
7	7	2	-	P13	I/O	General purpose digital input and output pins
				UART0_RTS	O	UART0 RTS pin
				I2C0_SDA	I/O	I2C0 SDA pin

				PWM_CH3	O PWM	channel 3 output pin
				SPI1_CS	I/O	SPI1 CS pin
				SPI0_CS	I/O	SPI0 CS pin
				USB_DM	AI/AO USB DM pin	
8	8	3	-	P14	I/O General	purpose digital input and output pins
				UART0_CTS	I	UART0 CTS pin
				I2C0_SCL	I/O	I2C0 SCL pin
				PWM_CH2	O PWM	channel 2 output pin
				SPI0_CLK	I/O	SPI0 clock pin
				USB_DP	AI/AO USB DP pinout	
9	9	-	-	P15	I/O General	purpose digital input and output pins
				SPI0_CS	I/O	SPI0 CS pin
				PWM_CH5	O PWM	channel 5 output pin
				TM0_EXT	I	TIMER0 external input pin
				UART0_RX	I	UART0 RX pin
10	-	-	-	P30	I/O General	purpose digital input and output pins
				UART1_RX	I	UART1 RX pin
				PWM_CH6	O PWM	channel 6 output pin
				SPI1_MISO	I/O	SPI1 MISO pin
11	10	-	-	P16	I/O General	purpose digital input and output pins
				UART0_TX	O	UART0 TX pin
				PWM_CH0	O PWM	channel 0 output pin
				SPI1_CLK	I/O	SPI1 clock pin
				I2C0_SCL	I/O	I2C0 SCL pin
				TIMER2_CNT_OUT O		TIMER2 output pin
12	11	-	15	P17	I/O General	purpose digital input and output pins
				UART0_RX	I	UART0 RX pin
				TM2_EXT	I	TIMER2 external input pin
				PWM_CH1	O PWM	channel 1 output pin
				RCH	O	RCH output pin
13	13	4	-	NRSET	I Reset pin	
14	12	5	17 DVDD		P	HLDO output pin, typical value 1.2V
15	14	6	18 XC1		AO external 32MHz clock source output	

16	15	7	19	XC0	AI	external 32MHz clock source input
17	16	9	20	VCC_RF	P	RF power supply port, can be directly connected to the VOUT_BK pin
18	-	8	2	GND	P	Common ground (VSS)
19	-	-	-	GND	P	Common ground (VSS)
20	17	10	1	ANT	AI/AO	RF antenna pin, an external antenna is required when in use
twenty one	-	-	-	P31	I/O	General purpose digital input and output pins
				UART1_TX	O	UART1 TX pin
				PWM_CH7	O	PWM channel 7 output pin
				SPI1_MOSI	I/O	SPI1 MOSI pin
twenty two	18	11	3	P22	I/O	General purpose digital input and output pins
				SPI1_CLK	I/O	SPI1 clock pin
				PWM_CH0	O	PWM channel 0 output pin
				SPI0_MOSI	I/O	SPI0 MOSI pin
				ADC_CH2	AI	ADC Channel 2 Pin
twenty three	19	12	4	P23	I/O	General purpose digital input and output pins
				SPI1_CS	I/O	SPI1 CS pin
				PWM_CH1	O	PWM channel 1 output pin
				DPLL_DIV8	O	DPLL_DIV8 Output Pin
				I2C0_SCL	I/O	I2C0 SCL pin
				ADC_CH1	AI	ADC Channel 1 Pin
twenty four	20	13	6	P20	I/O	General purpose digital input and output pins
				XTL1	AO	external 32.768kHz clock source output
				SPI1_MISO	I/O	SPI1 MISO pin
				PWM_CH6	O	PWM channel 6 output pin
25	twenty one	14	7	P21	I/O	General purpose digital input and output pins
				XTL0	AI	external 32.768kHz clock source input
				SPI1_MOSI	I/O	SPI1 MOSI pin
				PWM_CH7	O	PWM channel 7 output pin
26	twenty two	-	5	P24	I/O	General purpose digital input and output pins
				UART1_RX	I	UART1 RX pin
				SPI1_MISO	I/O	SPI1 MISO pin
				PWM_CH2	O	PWM channel 2 output pin
				TIMER1_CNT_OUT	O	TIMER1 output pin

				ADC_CH6	AI	ADC Channel 6 Pin
27	-	-	9	P25	I/O	General purpose digital input and output pins
				UART1_TX	O	UART1 TX pin
				SPI1_MOSI	I/O	SPI1 MOSI pin
				PWM_CH3	O	PWM channel 3 output pin
				I2C0_SDA	I/O	I2C0 SDA pin
				ADC_CH7	AI	ADC Channel 7 Pin
28	-	-	-	P26	I/O	General purpose digital input and output pins
				UART1_RTS	O	UART1 RTS pin
				PWM_CH4	O	PWM channel 4 output pin
				SPI1_CS	I/O	SPI1 CS pin
29	-	-	-	P27	I/O	General purpose digital input and output pins
				UART1_CTS	I	UART1 CTS pin
				PWM_CH5	O	PWM channel 5 output pin
				SPI1_CLK	I/O	SPI1 clock pin
				XTH	O	XTH Output pin
30	twenty three	15	8	P00	I/O	General purpose digital input and output pins
				SWD_CLK	I	SWD clock input pin
				UART1_RX	I	UART1 RX pin
				I2C0_SCL	I/O	I2C0 SCL pin
				SPI0_CLK	I/O	SPI0 clock pin
31	twenty four	16	10	P01	I/O	General purpose digital input and output pins
				SWD_DAT	I/O	SWD data input and output pin
				UART1_TX	O	UART1 TX pin
				I2C0_SDA	I/O	I2C0 SDA pin
				SPI0_CS	I/O	SPI0 CS pin
32	25	-	-	P02	I/O	General purpose digital input and output pins
				UART0_CTS	I	Uart0 CTS pin
				SPI1_MISO	I/O	SPI1 MISO pin
				PWM_CH3	O	PWM channel 3 output pin
				UART1_CTS	I	UART1 CTS pin
33	26	-	-	P03	I/O	General purpose digital input and output pins
				UART0_CTS	I	UART0 CTS pin

				SPI0_CS	I/O	SPI0 CS pin
				PWM_CH2	O	PWM channel 2 output pin
				TIMER0_CNT_OUT	O	TIMER0 output pin
34	27	-	-	P04	I/O	General purpose digital input and output pins
				UART0_RTS	O	UART0 RTS pin
				SPI0_CLK	I/O	SPI0 clock pin
				PWM_CH3	O	PWM channel 3 output pin
				TM0_EXT	I	TIMER0 external input pin
				ADC_CH0	AI	ADC Channel 0 Pin
35	28	-	-	P05	I/O	General purpose digital input and output pins
				UART0_TX	O	UART0 TX pin
				SPI0_MISO	I/O	SPI0 MISO pin
				PWM_CH4	O	PWM channel 4 output pin
36	29	17	-	VDD_FLASH	P	FLASH power input pin
37	30	-	-	P06	I/O	General purpose digital input and output pins
				UART0_RX	I	UART0 RX pin
				SPI0_MISO	I/O	SPI0 MISO pin
				PWM_CH5	O	PWM channel 5 output pin
				TM1_EXT	I	TIMER1 external input pin
				AHB_CLK	O	AHB_CLK output pin
38	31	18	11	DVSS	P	Common ground terminal of DCDC power supply, independent power ground
39				VSS_BK		
40	32	19	12	VOUT_BK	P	DCDC voltage output pin
41	33	twenty one	-	E-PAD	P	Chip bottom pad, common ground terminal

5 Electrical characteristics

Maximum and minimum values

The notes below each table indicate that the data are obtained through comprehensive evaluation, design simulation and/or process characterization.

The test will be carried out on the production line; based on the comprehensive evaluation, the minimum and maximum values are obtained after the sample test.

The mean value is obtained by adding or subtracting three times the standard distribution (mean ± 3ÿ).

5.1 RF Characteristics

Table 5-1 General RF characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
f	Operating frequency		2400	-	2483 MHz	
PLLres	PLL Programming		244	1M	-	Hz
DR	Resolution Bit Rate		0.125	-	2	Mbps
ÿfBLE,2M	BLE mode 2Mbps modulation frequency deviation		450	500	550	kHz
ÿfBLE,1M	BLE mode 1Mbps modulation frequency deviation		225	250	275	kHz
ÿf297,2M	297 Mode 2Mbps Modulation Frequency Deviation		450	500	550	kHz
ÿf297,1M	297 Mode 1Mbps Modulation Frequency Deviation		225	250	275	kHz
ÿfN,2M	N mode 2Mbps modulation frequency deviation		-	320	-	kHz
ÿfN,1M	N mode 1Mbps modulation frequency deviation		-	170	-	kHz
fBLE,CS,2M	BLE mode 2Mbps channel spacing		-	2	-	MHz
fBLE,CS,1M	BLE mode 1Mbps channel spacing		-	2	-	MHz
f297,CS,2M	297 mode 2Mbps channel spacing		-	2	-	MHz
f297,CS,1M	297 mode 1Mbps channel spacing		-	1	-	MHz
fN,CS,2M	N mode 2Mbps channel spacing		-	2	-	MHz
fN,CS,1M	N mode 1Mbps channel spacing		-	1	-	MHz

Table 5-2 TX characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
PRFTX	output power		-	-	9	dBm
PRFC	power control range		-	40	-	dB
PRFCR	Power Stepping		-	-	±3	dB
PRF1M, 1	First adjacent channel leakage ratio@1Mbps		-	-40	-	dBc
PRF1M,2	Second adjacent channel leakage ratio @1Mbps		-	-56	-	dBc
PRF1M, ÿ3	Third adjacent channel leakage ratio@1Mbps		-	-60	-	dBc

PRF2M,2	First Adjacent Channel Leakage Ratio@2Mbps		-	-37	-	dBc
PRF2M,4	Second Adjacent Channel Leakage Ratio@2Mbps		-	-58	-	dBc
PRF2M,ÿ6M	Third adjacent channel leakage ratio@2Mbps		-	-60	-	dBc
PBW1M	20dB bandwidth@1Mbps		-	1.3	-	MHz
PBW2M	20dB bandwidth @2Mbps		-	2.3	-	MHz
PSP,1	Stray power @ÿ1GHz		-	-	-73 dBm	
PSP,2	Spurious power @ ÿ 1GHz		-	-	-49 dBm	

Table 5-3 RX characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
PRX,MIX	Receive maximum input power		-	0	-	dBm
PSENS,1M,BLE	BLE mode 1Mbps receiving sensitivity		-	-96	-	dBm
PSENS,2M,BLE	BLE mode 2Mbps receiving sensitivity		-	-93	-	dBm
PSENS,125K,BLE	BLE mode 125kbps receiving sensitivity		-	-99	-	dBm
PSENS,500K,BLE	BLE mode 500kbps receiving sensitivity		-	-99	-	dBm
PSENS,250K,B	B mode 250kbps receiving sensitivity		-	-99	-	dBm
PSENS,1M,297	297 Mode 1Mbps Receiving Sensitivity	Sensitivity, 1Mbps ideal transmitter, ÿ37 bytes, Bit Error Rate = 0.1%	-	-93	-	dBm
PSENS,2M,297	297 Mode 2Mbps Receiving Sensitivity		-	-91	-	dBm
PSENS,250K,297	297 Mode 250kbps Receiving Sensitivity		-	-99	-	dBm
PSENS,1M,N	N mode 1Mbps receiving sensitivity		-	-93	-	dBm
PSENS,2M,N	N mode 2Mbps receiving sensitivity		-	-91	-	dBm
PSENS,250K,N	N mode 250kbps receiving sensitivity		-	-99	-	dBm
C/I CO,1M,BLE	Same-channel interference suppression@1Mbps		-	6	-	dB
C/I 1M,1M,BLE	Interval 1M Adjacent channel selectivity @1Mbps		-	-3	-	dB
C/I 2M,1M,BLE	Interval 2M Adjacent channel selectivity @1Mbps		-	-40	-	dB
C/I ÿ3M,1M,BLE	Adjacent channel selectivity above 3M @1Mbps		-	-45	-	dB
C/I Image,1M,BLE	Mirror selectivity@1Mbps		-	-twenty three	-	dB
C/I Image±1M,1M,BLE	Mirror ±1M Selective@1Mbps		-	-38	-	dB
C/I ÿ6M,1M,BLE	Adjacent channel selectivity above 6M @1Mbps		-	-45	-	dB
C/I CO,2M,BLE	Same-channel interference suppression @2Mbps		-	7	-	dB
C/I 2M,2M,BLE	Interval 2M Adjacent channel selectivity @2Mbps		-	-3	-	dB
C/I 4M,2M,BLE	Interval 4M Adjacent channel selectivity @2Mbps		-	-37	-	dB

C/I ÿ6M,2M,BLE	Adjacent channel selectivity above 6M @2Mbps		-	-35	-	dB
C/I Image,2M,BLE	Mirror Selectivity@2Mbps		-	-twenty three	-	dB
C/I Image±2M,2M,BLE	Mirror ±2M Selective@2Mbps		-	-36	-	dB
C/I ÿ12M,2M,BLE	Adjacent channel selectivity above 12M @2Mbps		-	-35	-	dB
C/I CO,125K,BLE	Same channel interference suppression @125kbps		-	-6	-	dB
C/I 1M,125K,BLE	Interval 1M Adjacent channel selectivity @125kbps		-	-13	-	dB
C/I 2M,125K,BLE	Interval 2M Adjacent channel selectivity @125kbps		-	-37	-	dB
C/I ÿ3M,125K,BLE	Adjacent channel selectivity above 3M @125kbps		-	-49	-	dB
C/I Image,125K,BLE	Mirror selectivity @125kbps		-	-31	-	dB
C/I Image±1M,125K,BLE	Mirror ±1M Selective@125kbps		-	-48	-	dB
C/I CO,500K,BLE	Same-channel interference suppression @500kbps		-	1	-	dB
C/I 1M,500K,BLE	Interval 1M Adjacent channel selectivity @500kbps		-	-6	-	dB
C/I 2M,500K,BLE	Interval 2M Adjacent channel selectivity @500kbps		-	-31	-	dB
C/I ÿ3M,500K,BLE	Adjacent channel selectivity above 3M @500kbps		-	-43	-	dB
C/I Image,500K,BLE	Mirror selectivity @500kbps		-	-25	-	dB
C/I Image±1M,500K,BLE	Mirror ±1M Selective@500kbps		-	-41	-	dB

Table 5-4 RSSI characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
RSSIRFC	RSSI indication range		-90	-	-15	dBm
RSSIAuu	RSSI Accuracy		-	±2	-	dB
RSSIRes	RSSI resolution		-	0.25	-	dB
RSSIPer	RSSI sampling period		-	0.25	-	us

Table 5-5 RF Timing Characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
TRX-TX	TX and RX switching time		-	150	-	us

Table 5-6 RF power consumption characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
ITX,P9dBm,DCDC	9dBm power output current @DC-DC		-	13.97	-	mA
ITX,P6dBm,DCDC	6dBm power output current @DC-DC		-	9.59	-	mA
ITX,P4dBm,DCDC	4dBm power output current @DC-DC		-	6.67	-	mA
ITX,P0dBm,DCDC	0dBm power output current @DC-DC		-	5.06	-	mA
ITX,P-4dBm,DCDC	-4dBm power output current @DC-DC		-	2.56	-	mA
ITX,P-8dBm,DCDC	-8dBm power output current @DC-DC		-	2.15	-	mA
ITX,P-12dBm,DCDC	-12dBm power output current @DC-DC		-	1.87	-	mA
ITX,P-16dBm,DCDC	-16dBm power output current @DC-DC		-	1.64	-	mA
ITX,P-20dBm,DCDC	-20dBm power output current @DC-DC		-	1.53	-	mA
ITX,P-40dBm,DCDC	-40dBm power output current @DC-DC		-	1.12	-	mA
ITX,P9dBm,LDO	9dBm power output current @ LDO		-	35.55	-	mA
ITX,P6dBm,LDO	6dBm power output current @ LDO		-	16.28	-	mA
ITX,P4dBm,LDO	4dBm power output current @ LDO		-	12.1	-	mA
ITX,P0dBm,LDO	0dBm power output current @ LDO		-	9.43	-	mA
ITX,P-4dBm,LDO	-4dBm power output current @ LDO		-	5.74	-	mA
ITX,P-8dBm,LDO	-8dBm power output current @LDO		-	5.08	-	mA
ITX,P-12dBm,LDO	-12dBm power output current @ LDO		-	4.62	-	mA
ITX,P-16dBm,LDO	-16dBm power output current @ LDO		-	4.04	-	mA
ITX,P-20dBm,LDO	-20dBm power output current @LDO		-	3.92	-	mA
ITX,P-40dBm,LDO	-40dBm power output current @ LDO		-	3.00	-	mA
IRX,1M,DCDC	RX 1Mbps Current@DC-DC		-	2.50	-	mA
IRX,2M,DCDC	RX 2Mbps Current@DC-DC		-	2.71	-	mA
IRX,1M,LDO	RX 1Mbps Current@LDO		-	4.82	-	mA
IRX,2M,LDO	RX 2Mbps Current@LDO		-	5.67	-	mA

Test conditions and methods:

1. 1M mode uses BLE ADV broadcast mode to test the transmit and receive power consumption
2. 2M mode uses the power consumption when BLE is connected
3. The power consumption tested is the RF peak power consumption
4. The test method uses the total power consumption minus the power consumption of the MCU when RF is not working to calculate the final power consumption
5. The sample software tested is based on peripheral_hr
6. 3.3V power supply

5.2 GPIO Features

Table 5-7 Single IO combination test

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VIH	input high level threshold voltage	TA=25°C load	0.7*VDD -		VDD	V
VIL	Input Low Level Threshold Voltage	capacitance=20pF, TA=25°C	VSS	-	0.2*VDD	V
VHYS	Input hysteresis voltage, Vhys=VIH-VIL	TA=25°C	-	-	0.3*VDD	V
CIana	simulates input capacitance	TA=25°	-	300	-	f
ILkg	Leakage current, open drain mode or input Enter Mode	VDDÿVINÿ3.6V -		-	TBD	uA
RPU	pull-up resistor	Vin = VSS, VDD =3.3V	-	50	-	kÿ
RPD	pull-down resistor	Vin = VSS, VDD =3.3V	-	100	-	kÿ
VI	Input voltage	TA=25°C	VSS	-	VDD	V
VO	input voltage	TA=25°C	VSS	-	VDD	V
Isource	Single pin source current (push-pull input Output) (Except P03, P15)	Vin = VDD-0.5V	8.5	9.5	10.5	mA
	Single pin source current (push-pull input (out) (P03, P15)		twenty one	twenty two	twenty three	mA
ISink	Single pin sink current (push-pull input out)	Vin = VSS + 0.5V, TA = 25°C	19.5	20.5	21.5	mA
fPort_CLK	IO output frequency	Load capacitance = 20pF -		-	48	MHz

Table 5-8 Combination test

Description	Conditions	Status	Remark
After power on, IO default state	VDD=3.3V , TA=25°C	P00, P01: Pull-up input state Others: High impedance	
IO status in sleep mode	VDD=3.3V , TA=25°C	Deepsleep: All GPIOs remain Standby_m1: All GPIOs remain Standby_m0: P00, P01, P02 hold	
IO status during reset	VDD=3.3V ,TA=25°C	P00, P01: Pull-up input state Others: High impedance	

Table 5-9 nRESET input characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
VILR negative threshold voltage, nRESET VDD=1.8V-3.3V ,TA=25°C VIHR			-	-	0.22*VDD V	
positive threshold voltage, nRESET VDD=1.8V-3.3V ,TA=25°C 0.48*VDD - Vhys_rst Schmitt trigger					-	V
voltage hysteresis VDD=1.8V-3.3V ,TA=25°C nRESET pin internal pull-up resistor			-	-	0.26*VDD V	
	VDD=3.3V ,TA=25°C RRST nRESET pin input filter pulse		-	7.5	-	kÿ
tFR, 0.3pF	time	VDD=3.3V ,TA=25°C	-	TBD -		ns

5.3 Reset Characteristics

Table 5-10 Reset characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VBOD	BOD undervoltage detection voltage	BODSEL<2:0> = 000(falling edge)ÿ dVDD/dtÿ3V/s	-	1.84	-	V
		BODSEL<2:0> = 001(falling edge)ÿ dVDD/dtÿ3V/s	-	1.96	-	
		BODSEL<2:0> = 010(falling edge)ÿ dVDD/dtÿ3V/s	-	2.05	-	
		BODSEL<2:0> = 011(falling edge)ÿ dVDD/dtÿ3V/s	-	2.16	-	
		BODSEL<2:0> = 100(falling edge)ÿ dVDD/dtÿ3V/s	-	2.25	-	
		BODSEL<2:0> = 101(falling edge)ÿ dVDD/dtÿ3V/s	-	2.35	-	
		BODSEL<2:0> = 110(falling edge)ÿ dVDD/dtÿ3V/s	-	2.47	-	
VBODhys BOD hysteresis voltage dVDD/dtÿ3V/s			100	-	160 mV	
TBOD_RE1	BOD response time, normal model	dVDD/dtÿ3V/s	1	32	32	Slow clock
IBOD	BOD working current dVDD/dtÿ3V/s		-	0.5	-	uA
VPOR	POR Brown-out Detection Voltage	Rising edge, dVDD/dtÿ3V/s	-	1.71	-	V
		Falling edge, dVDD/dtÿ3V/s	-	1.69	-	V
TPOR	POR power-on ratio VBAT Delay	VDD =3.3V	-	1.12	-	ms
VLVR	LVR detection voltage	Falling edge, dVDD/dtÿ3V/s	-	1.85	-	V
TLVR_RE LVR response time		TA=25°C, dVDD/dtÿ3V/s	1	32	32	Slow clock
ILVR	LVR operating current	TA=25°C, dVDD/dtÿ3V/s	-	0.1	-	uA

5.4 Clock Characteristics

Table 5-11 HXTAL characteristics

Symbol	Description	Conditions	Parameter Min	Typ	Max	Unit
HkDJ	High speed crystal oscillator (HXTAL) frequency	VDD=3.3V ,TA=25°C	-	32	-	MHz
CLoadHXTL	Crystal load capacitance	VDD=3.3V ,TA=25°C	7	9	12	pF
IDX	HXTAL oscillator operating current	VDD=3.3V ,TA=25°C	-	250	-	ÿA
tX	HXTAL oscillator start-up time	VDD=3.3V ,TA=25°C, ESR=40ÿ, CHXTL= 9pF	-	465	-	ÿs
tSUHXTL	Quick HXTAL Oscillator quick start time	VDD=3.3V ,TA=25°C, ESR=40ÿ, CHXTL= 9pF	-	155	-	ÿs
ESRHXTL	equivalent series resistance	VDD=3.3V ,TA=25°C	-	40	-	ÿ
Frequency tolerance of FTOLHXTL crystal		VDD=3.3V ,TA=25°C	-20	-	20	ppm
PDHXTL	Excitation Power	VDD=3.3V ,TA=25°C	-	-	100 ÿW	

Table 5-12 LXTAL characteristics

Symbol	Description	Conditions	Parameter Min	Type	Max	Unit
FWf	Low speed crystal oscillator (LXTAL) frequency	VDD=3.3V ,TA=25°C -		32.768	-	kHz
IDDLXTL	LXTAL Oscillator Operating Current	VDD=3.3V ,TA=25°C 0.3 LXTAL		0.45	0.67	ÿA
tXTL	oscillator normal start-up time VDD=3.3V ,TA=25°C 330			550	1500 ms	
tSULXTL Quick LXTAL Oscillator quick start time VDD = 3.3V , TA = 25 ° C - ESRLXTL				125	-	ms
equivalent series resistance		VDD=3.3V ,TA=25°C -		70	-	kÿ
CLoadLXTL	crystal load capacitance	VDD=3.3V ,TA=25°C -		7	-	pF
FTOLLXTL	crystal frequency tolerance	VDD=3.3V ,TA=25°C -20		-	20	ppm
PDLXTL	Excitation Power	VDD=3.3V ,TA=25°C -		-	1	ÿW

Table 5-13 32MHz RCH characteristics

Symbol	Description	Conditions	Parameter Min	Type	Max	Unit
fIRC32M	oscillator frequency	VDD=3.3V ,TA=25°C	-	32	-	MHz
ACCIRC32M	Frequency Accuracy	VDD=3.3V, TA=-40°C ~+105°C - VDD=3.3V,		-	-	%
		TA=-20°C ~ +85°C - VDD=3.3V, TA=25°C		-	-	%
			-	±1	-	%
DIRC32M	IRC32M Oscillator Duty Cycle	VDD=3.3V, fIRC32M=32MHz, TA=25°C	48	50	52 %	
IDDIRC32M	operating current	VDD=3.3V, fIRC32M=32MHz, TA=25°C	-	82	-	ÿA
tSUIRC32M	stabilization time	VDD=3.3V, fIRC32M=32MHz, TA=25°C	-	5	-	ÿs

dfIRC32M	25°C, frequency supply voltage drift shift	VDD=1.8V~3.6V, TA=25°C	-	0.5	-	%/V

Table 5-14 32kHz RCL characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
fIRC32k	oscillator frequency	VDD=3.3V ,TA=25°C	-	32	-	kHz
ACCIRC32K	Frequency Accuracy	VDD=3.3V, TA=25°C (After calibration)	-	±500	-	ppm
DIRC32K	IRC32K Oscillator Duty Cycle	VDD=3.3V, fIRC32K=32kHz, TA=25°C	48	50	52	%
IDDIRC32K	operating current	VDD=3.3V, fIRC32K=32kHz, TA=25°C	-	310	-	nA
tSUIRC32K	startup time	VDD=3.3V, fIRC32K=32kHz, TA=25°C	-	480	-	ÿs
dfIRC32K	25°C, frequency supply voltage drift	VDD=1.8V~3.6V, TA=25°C	-	1	-	%/V

Table 5-15 DPLL characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
f	PLL input clock frequency	VDD=3.3V, TA=25°C	-	32	-	MHz
fPLL	PLL output clock frequency	VDD=3.3V, TA=25°C	48	48	48	MHz
IPLL	operating current	VDD=3.3V, TA=25°C	96	105	145	ÿA

5.5 ADC Characteristics

Table 5-16 Power supply and input range conditions

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
VAx(VBG adc)	Analog input voltage range, VBG (1.2V) gear	VDD=3.3V, TA=25°C	0	-	1.2	V
VAx(VDD)	analog input voltage range, VDD range	VDD=3.3V, TA=25°C	0	-	VDD	V
IADC	ADC supply current	VDD=3.3V, TA=25°C Fadc=16MHz	-	0.55	-	mA
Csample	Internal sample and hold capacitors (not included) Including PAD and PCB capacitors)		-	10	-	pF
RADC	sampling switch resistance	0V ÿ VAx ÿ VDD	-	300	-	ÿ
RIn	External input impedance, continuous sampling	0V ÿ VAx ÿ VDD	0.86	-	4734.70 kÿ	

Table 5-17 ADC built-in voltage reference

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
VBG	ADC has a built-in 1.2V reference voltage	VDD=3.3V, TA=25°C	1.1	1.2	1.3	V
TCoef	Temperature coefficient	TA=-40ÿ~105ÿ; VDD=1.8V~3.6V	-	30	-	ppm/ÿ

Table 5-18 Time parameters

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
FADC	ADC clock frequency	VDD=3.3V, TA=25°C	4	16	twenty four	MHz
TS	Sampling time	VDD=3.3V, TA=25°C	4	1539	8192	1/Fadc
TCONV	conversion time	VDD=3.3V, TA=25°C	32	1580	8298	1/Fadc

Table 5-19 Linearity parameters

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
INL	Integral linearity error	VDD=3.3V, TA=25°C	-	-	±3	LSB
DNL	Differential linearity error	VDD=3.3V, TA=25°C -		-	±2	LSB
SNR	Signal to Noise Ratio	Fadc = 16MHz Input clock 250kHz VDD=3.3V, TA=25°C	-	64.3	-	dB
THD	Total Harmonic Distortion		-	75	-	dB
SFDR	Spurious Free Dynamic Range		-	77.29	-	dB
ENOB	Effective Bits		-	10.33	-	Bit

Table 5-20 RIN

ADC significant bit	FADC(MHz)	Ts(cycles)	Ts(us)	Rinmax(kÿ)
12	32	4	0.125	0.86
12	32	8	0.25	2.01
12	32	32	1	8.95
12	32	64	2	18.20
12	32	128	4	36.69
12	32	8192	256	2367.20
12	16	4	0.25	2.01
12	16	8	0.5	4.32
12	16	32	2	18.20
12	16	64	4	36.69
12	16	128	8	73.68
12	16	8192	512	4734.70

Note: The sampling condition is continuous sampling.

5.6 PMU Characteristics

Table 5-21 PMU characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
DVDD	DVDD output voltage range, external capacitor	VDD=3.3V, TA=25°C	1.1	1.2	1.4	V
VDD_FLASH*	VDD_FLASH output voltage range, external capacitor capacitance	VDD=3.3V, TA=25°C	1.8	-	VDD	V

*Note: VDD_FLASH output is 1.8V or VDD

5.7 General working conditions

Table 5-22 General working conditions

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VDD*	Operating voltage (DCDC-OFF mode)	TA=25°C	1.8	-	3.6	V
	Operating voltage (DCDC-ON mode)		2.0	-	3.6	V
TST	Storage temperature	-	-65	-	150	ÿ
TA	Ambient temperature	-	-40	-	105	ÿ
TJ-QFN32	Junction temperature	QFN32 4x4x0.75-P0.4	-40	-	125	ÿ
RÿJA-QFN32	Thermal Resistance	QFN32 4x4x0.75-P0.4	-	41	-	ÿ/W

*Note: VDD is the input voltage of the chip power pin VBAT

5.8 DCDC Characteristics

Table 5-23 DCDC characteristics

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VIN_DCDC	input voltage range	VDD=3.3V, TA=25°C	2	-	3.6	V
VOUT_DCDC	output voltage range	VDD=3.3V, TA=25°C	1.2	1.5	2	V
TEN_DCDC	start time	VDD=3.3V, ILOAD=10mA, TA=25°C, LDCR=80mÿ	-	200	-	us
ÿ	efficiency	VDD=3.3V, ILOAD=10mA, TA=25°C, LDCR=80mÿ	-	88	-	%
VRPLDCDC	Ripple	VDD=3.3V, TA=25°C	-	43	-	mv
IOUT	drive peak current	VDD=3.3V, TA=25°C	-	100	-	mA
IAVG	drive average current	VDD=3.3V, TA=25°C	-	30	-	mA
LDCDC	effective inductance	VDD=3.3V, TA=25°C	-	2.2	-	ÿH
COUT_DCDC	Effective Load Capacitance	VDD=3.3V, TA=25°C	1	4.7	-	ÿF
Fosc_DCDC	oscillation frequency	VDD=3.3V, TA=25°C	-	-	1000	kHz

5.9 Electrical sensitivity

Table 5-24 Electrical sensitivity

Symbol	Description	Conditions	Parameter			Unit
			Min	Type	Max	
VESDHBM[1]	ESD @ Human Body Mode	TA=25°C	-	±4	-	kV
VESDCDM[2]	ESD @ Charge Device Mode	TA=25°C	-	±2000	-	V
VESDMM[3]	ESD @ machine Mode	TA=25°C	-	±200	-	V
Ilatchup[4]	Latch up current Note:	TA=25°C	-	±500	-	mA

1. Determined according to ANSI/ESDA/JEDEC JS-001 standard, Electrostatic Discharge Sensitivity Test - Human Body Model (HBM) - Device Level
2. Determined according to ANSI/ESDA/JEDEC JS-002 Electrostatic Discharge Sensitivity (ESD) test standard
3. Determined according to JESD22-A115-C Electrostatic Discharge Sensitivity (ESD) test standard
4. Measured according to JEDEC EIA/JESD78 standard

5.10 Absolute Maximum Ratings

Table 5-25 Absolute maximum ratings

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VDD[1]	External main supply voltage	TA=25°C	-0.3	-	3.6	V
VCC_RF	chip RF circuit input voltage	TA=25°C	-0.3	-	3.6	V
VIN	Input voltage on other pins	TA = 25°C	VSS[2]	-	VDD	V
PVDD	Extreme power consumption	VDD=3.3V, TA=25°C DCDC power supply	-	-	TBD	mW

Note:

1. VDD is the input voltage of the chip power pin VBAT
2. VSS is GND

5.11 MCU Current Characteristics

Symbol	Parameter	Conditions	Clock (Hz)	DCDC OFF Typ(mA)	DCDC ON Typ(mA)
Run mode	All peripherals clock on, run while(1) in flash	System clock source: RCH (.cal 32M) [1]	4M	0.91	0.686
			8M	1.3	0.916
			16M	2.09	1.36
			32M	3.77	2.28
		System clock source: XTH (.off rch) [2]	4M	1.42	0.985
			8M	1.81	1.2
			16M	2.61	1.64
			32M	4.29	2.54
		System clock source: DPLL (.ref rch(.cal 32M)) (.base 48M) [3]	6M	1.56	1.05
			12M	2.15	1.38
			24M	3.4	2.05
			48M	5.71	3.44
		System clock source: DPLL (.ref xth) (.base 48M) (.off rch) [4]	6M	2.08	1.34
			12M	2.69	1.67
			24M	3.95	2.34
			48M	6.18	3.57
		System clock source: DPLL (.ref rch(.cal 32M)) (.base 64M) [5]	4M	1.39	0.953
			8M	1.78	1.17
			16M	2.58	1.61
			32M	4.27	2.51
		System clock source: DPLL (.ref xth) (.base 64M) (.off rch) [6]	4M	1.9	1.25
			8M	2.3	1.46
			16M	3.13	1.91
			32M	4.85	2.82
	All peripherals clock off, run while(1) in flash	System clock source: RCH (.cal 32M) [1]	4M	0.736	0.523
			8M	0.943	0.719
			16M	1.37	0.963
			32M	2.23	1.43
		System clock source: XTH (.off rch) [2]	4M	0.945	0.723
			8M	1.15	0.84
			16M	1.57	1.07
			32M	2.42	1.54
		System clock source: DPLL (.ref rch(.cal 32M)) (.base 48M) [3]	6M	1.01	0.747
			12M	1.32	0.921
			24M	1.96	1.25
			48M	3.29	2.00
		System clock source: DPLL (.ref xth) (.base 48M) (.off rch) [4]	6M	1.22	0.903
			12M	1.54	1.05
			24M	2.19	1.41
			48M	3.54	2.12
		System clock source: DPLL	4M	0.925	0.693
			8M	1.14	0.818

		(.ref rch(.cal 32M)) (.base 64M) [5]	16M	1.56	1.05
			32M	2.43	1.53
		System clock source: DPLL (.ref xth) (.base 64M) (.off rch) [6]	4M	1.14	0.831
			8M	1.35	0.947
			16M	1.78	1.18
			32M	2.66	1.66

Note: Test conditions: DVDD=1.2V, VDD(VBAT)=3.3V, TA=25°C

Note:

1. RCH is calibrated to 32M and then divided.
2. Use XTH as the clock source, turn off RCH, and then divide the frequency.
3. Use RCH as the clock source of DPLL, calibrate RCH to 32M, set DPLL to 48M, and then divide the frequency.
4. Use XTH as the clock source of DPLL, set DPLL to 48M, turn off RCH, and then divide the frequency.
5. Use RCH as the clock source of DPLL, calibrate RCH to 32M, set DPLL to 64M, and then divide the frequency.
6. Use XTH as the clock source of DPLL, set DPLL to 64M, turn off RCH, and then divide the frequency.

Symbol	**Parameter**	**Conditions**		**Typ(ÿA)**
standby m0 wake by gpio				0.28
Symbol	**Parameter**	**Conditions**	**Clk source**	**Typ(ÿA)**
standby m1	wake by 32k timer	sw=0x0d [1]	XTL	1.88 [2]
		sw=0x1f [1]	XTL	2.18 [2]
	wake by gpio	sw=0x0d [1]		1.32 [2]
		sw=0x1f [1]		1.63 [2]
	wake by wdt	sw=0x0d [1]	XTL	1.85 [2]
		sw=0x1f [1]	XTL	2.18 [2]
	wake by lvr	sw=0x0d [1]		2.05 [2]
		sw=0x1f [1]		2.42 [2]
	wake by bod	sw=0x0d [1]		2.11 [2]
		sw=0x1f [1]		2.48 [2]
Deepsleep	wake by 32k timer	sw=0x0d [1]	XTL	2.96[2]
		sw=0x1f [1]	XTL	3.37[2]
	wake by gpio	sw=0x0d [1]		2.46[2]
		sw=0x1f [1]		2.86[2]
	wake by wdt	sw=0x0d [1]	XTL	2.99[2]
		sw=0x1f [1]	XTL	3.37[2]
	wake by peripheral timer	sw=0x0d [1]	XTL	4.03[2] [3]
		sw=0x1f [1]	XTL	4.73[2]
	wake by 32k timer*	sw=0x0d [1]	XTL	4.06[2] [3]
		sw=0x1f [1]	XTL	4.69[2]
Sleep	wake by gpio	LDO all peripheral clocks on LDO		4620
		non-essential peripheral clocks off DCDC all		1820
		peripheral clocks on DCDC non-		2700
		essential peripheral clocks off Note: Test conditions		1210

DVDD=1.2V, VDD(VBAT)=3.3V, TA=25°C, DCDC ON

*: PWM output enable

Note:

1. sw indicates the module control switch of PAN107 series power preservation in low power mode. The bit is 1 to preserve power, and the bit is 0 to cut off power.

   0d = sram32k+phy_sram+cpu_retention,1f = all_sram_retention

2. In LPLDOH Mode 2, the LPLDOH TRIM voltage has no effect and the effective voltage is the VREF_TRIM voltage.

   In this mode, LPLDOH undershoot will not occur and the bottom current will also decrease.

3. In Deepsleep mode, PWM can output waveforms normally, but only in deepsleep mode 2. For the description of deepsleep mode 2, refer to Note 4.

4. When register LP_FL_CTRL[31] is 1, the switch between LPLDOL and LPLDOH is closed, and LPLDOL is enabled.

   LPLDOL is powered by LPLDOH and is mode 2. The power consumption of mode 2 increases.

6 Reference Schematic

Please visit the following address to obtain the latest reference schematics and hardware design considerations: PAN10xx Hardware Reference Design Guide

7 Packaging Information

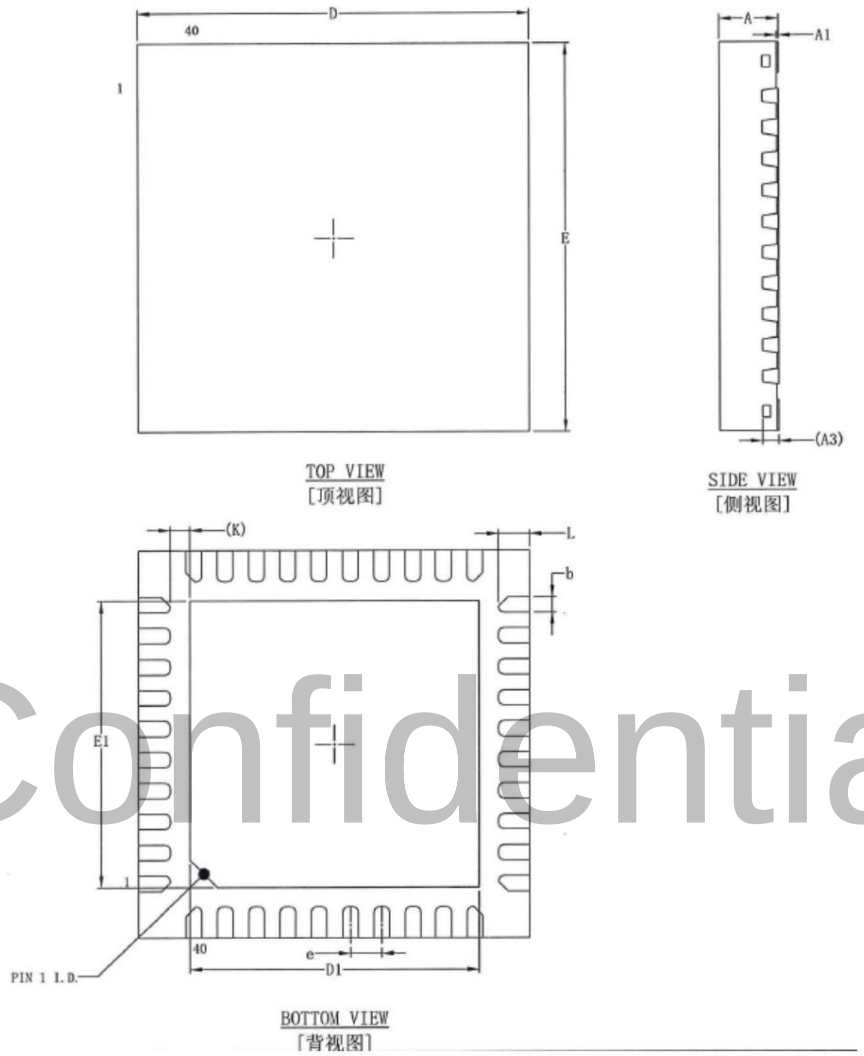


Figure 7-1 QFN40 package diagram

Table 7-1 QFN40 package dimensions

SYMBOL	MIN (mm)	NOM (mm)	MAX (mm)
A	0.70	0.75	0.80
A1	0.00	0.02	0.05
A3	0.203 REF		
b	0.15	0.20	0.25
D	4.90	5.00	5.10
D1	3.60	3.70	3.80
E	4.90	5.00	5.10
E1	3.60	3.70	3.80
e	0.40 BSC		
K	0.25 REF		
L	0.30	0.40	0.50

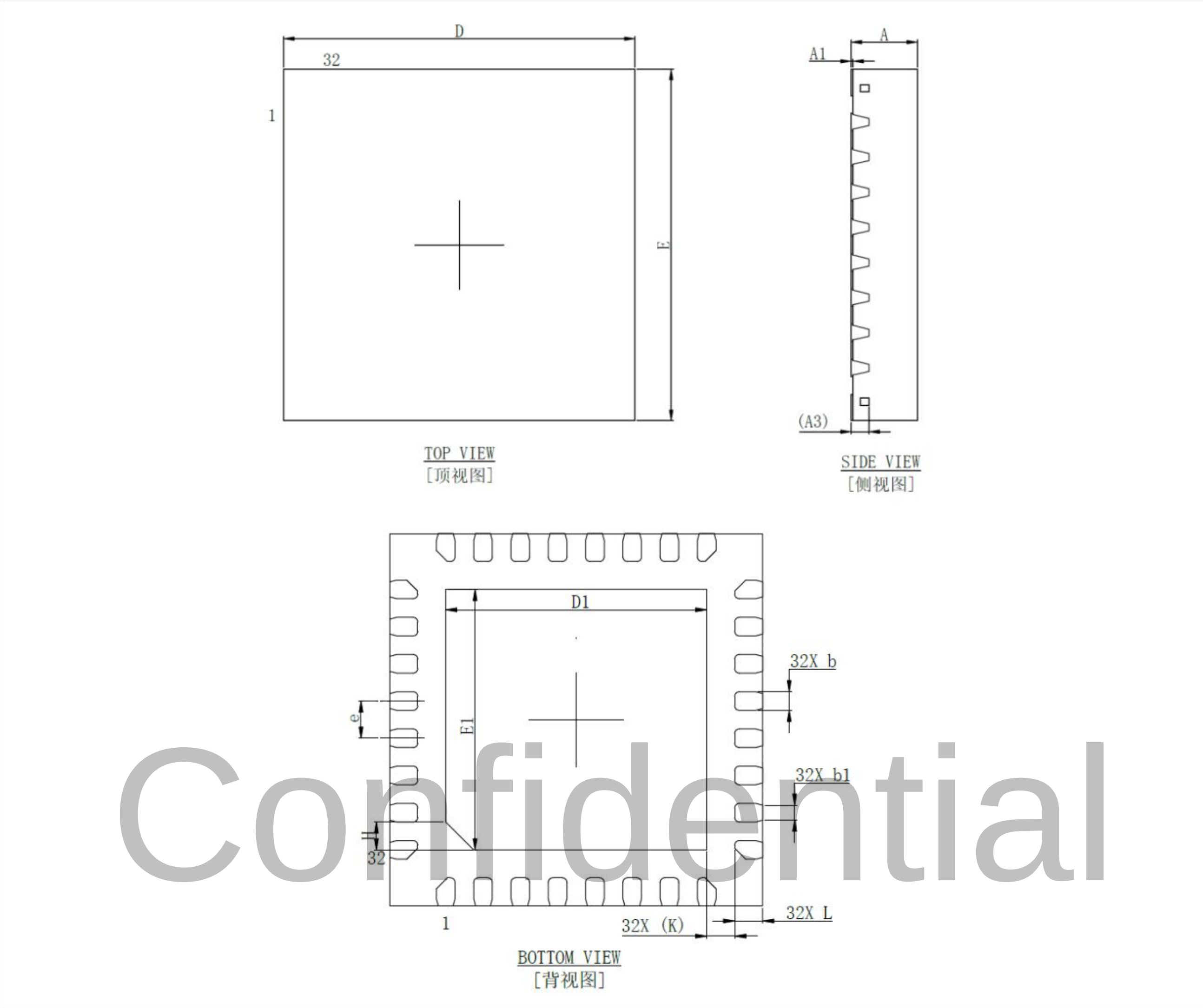


Figure 7-2 QFN32 package diagram

Table 7-2 QFN32 package dimensions

SYMBOL	MIN (mm)	NOM (mm)	MAX (mm)
A	0.70	0.75	0.80
A1	0.00	0.02	0.05
A3	0.203 REF		
b	0.15	0.20	0.25
b1	0.16 REF		
D	3.90	4.00	4.10
D1	2.65	-	2.90
E	3.90	4.00	4.10
E1	2.65	-	2.90
e	0.40 BSC		
K	0.30 REF		
L	0.20	-	0.40
H	0.30 REF		

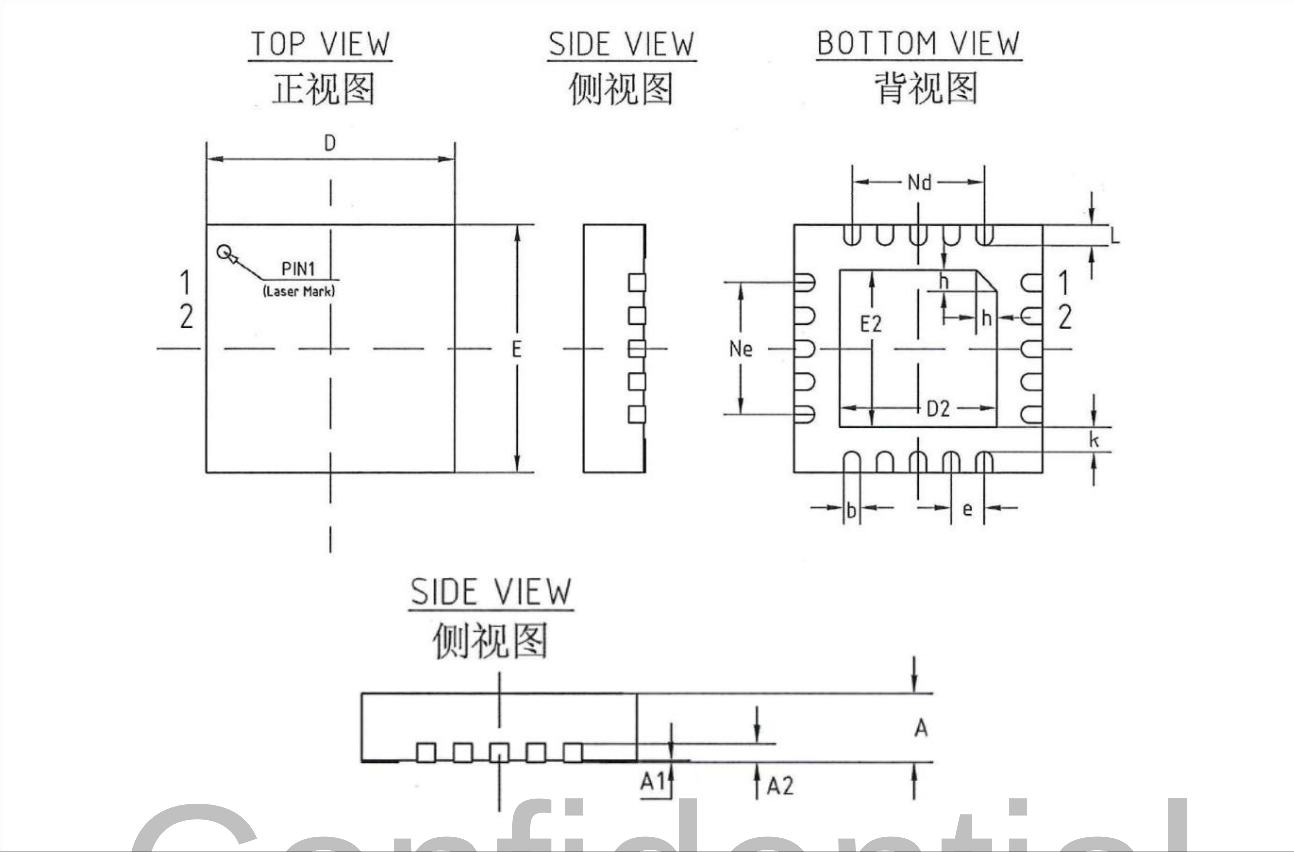


Figure 7-3 QFN20 package diagram

Table 7-3 QFN20 package dimensions

SYMBOL	MIN (mm)	NOM (mm)	MAX (mm)
A	0.70	0.75	0.80
A1	-	0.02	0.05
A3	0.203 REF		
b	0.15	0.20	0.25
D	2.90	3.00	3.10
D2	1.80	1.90	2.00
E	2.90	3.00	3.10
E2	1.80	1.90	2.00
e	0.40 BSC		
K	0.20	0.30	0.40
L	0.20	0.25	0.30
h	0.20	0.25	0.30
Ne	1.60 BSC		
Nd	1.60 BSC		

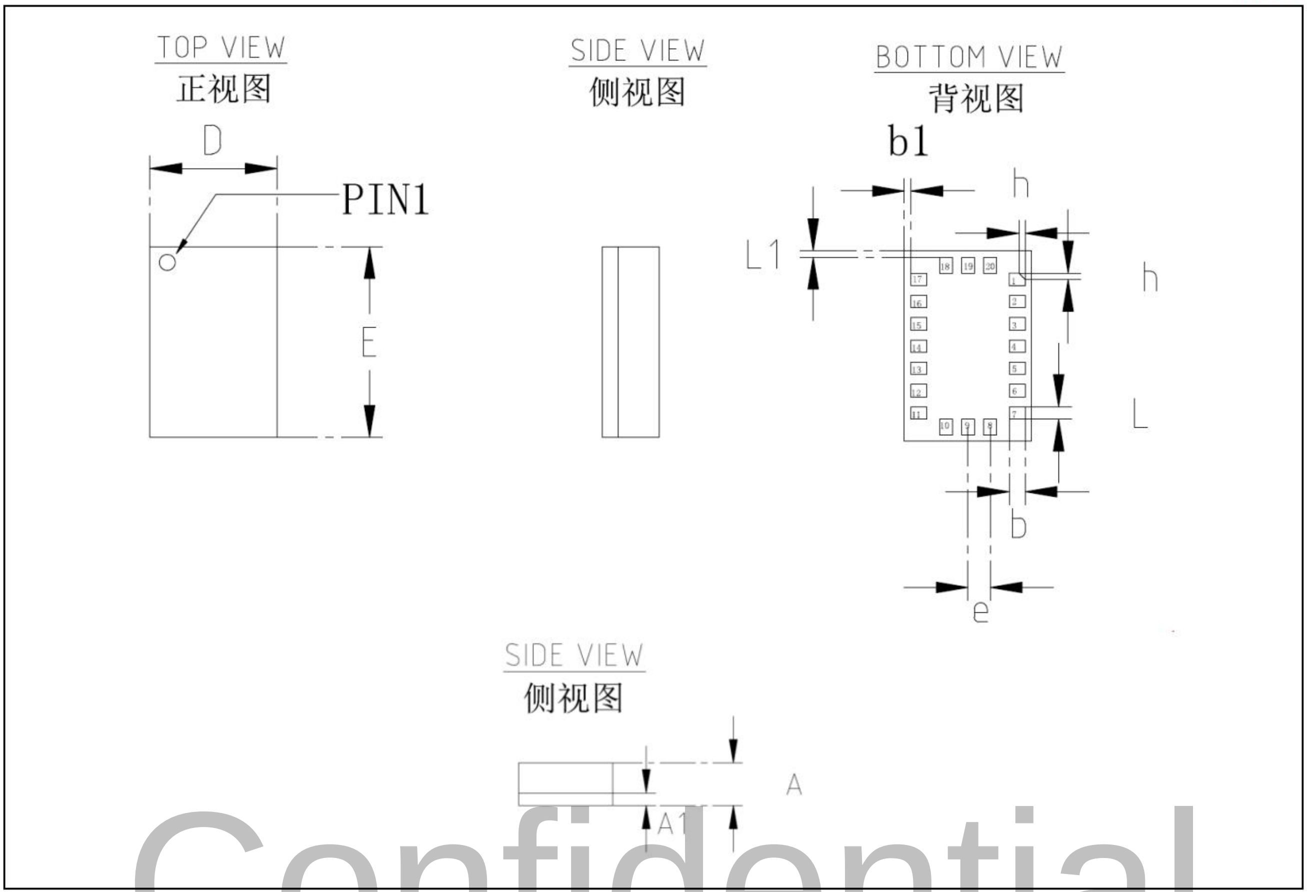


Figure 7-4 LGA20 package diagram

Table 7-4 LGA20 package dimensions

SYMBOL	MIN (mm)	NOM (mm)	MAX (mm)
A	0.825	0.900	0.975
A1	0.230	0.260	0.290
b	0.200	0.250	0.300
D	1.925	2.000	2.075
e	0.350BSC		
L1	0.100REF		
b1	0.100REF		
h	0.100REF		
E	2.925	3.000	3.075
L	0.150	0.200	0.250

Abbreviations

ACK Acknowledgement signal

ADC Analog-to-Digital Converter

bandgap bandgap reference

BLE Bluetooth Low Energy

BOD brownout detection

BQB Bluetooth Certification

Cache

CDM Charged Device Model

CS Chip Select

CTS Receive Request

DCDC Step-Down DC Converter

DMA Direct Memory Access

DPLL Digital Phase-Locked Loop

eFuse One-Time Programmable Memory

ESD Electrostatic Discharge

ext External IO port

ETSI European Telecommunications Standards Institute

FCC Federal Communications Commission

FMC Flash controller

GPIO General Purpose I/O.

HBM discharge pattern of a charged human body

HLDO Strong Drive Low Dropout Linear Regulator

I2C two-wire serial bus

IAP Application Programming

Keyscan

Latch-up

LDO Low Dropout Regulator

LED Light Emitting Diode

LVR Low Voltage Reset

MCU Microcontroller Unit

MISO Master In Slave Out

MM Discharge mode for powered machines

MOSI Master Out Slave In

PLL Phase Locked Loop

POR Power-On Reset

PWM Pulse Width Modulation

RAM Random Access Memory

RC Resistor Capacitor Crystal Oscillator

RF

RSSI Received Signal Strength

RX

SDA data signal line

SoC System on Chip

SPI Serial Peripheral Interface

SRAM Static Random Access Memory

SWD Serial Wire Debug

TEMP Temperature Sensor

TX emission

RTS Send Request

UART Universal Asynchronous Receiver/Transmitter

USB Universal Serial Bus

WDT Watchdog Timer

WWDT Serial Watchdog Timer

XTAL External crystal

Revision History

Version	Date	Content
1.0	Oct. 2023	First version released
1.1	Nov. 2023	Added reference schematic. Corrected typos of pin12 and pin13 in pin diagram. Corrected pin name of pin31 in pin diagram.
1.2	Dec. 2023	Updated pin information.
1.3	Jan. 2024	Added PAN1070UAEC.
1.4	Mar. 2024	Supplemented electrical characteristics.
1.5	Apr. 2024	Updated major features.
1.6	May. 2024	Updated electrical characteristics and pin diagram for QFN20.
1.7	Nov. 2024	Updated QFN32 package information, updated reference schematics, updated Table 5-7, and updated the typical value of standby m1 in MCU current characteristics.
1.8	Dec. 2024	Updated pin description and Table 5-7.
1.9	Mar. 2025	Added QFN40 and LGA20 packages.

Internal version, for reference only.

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